

Programming Fundamentals & Unit Testing

Course Introduction



SoftUni Team
Technical Trainers



SoftUni

Software University

<https://softuni.bg>

Have a Question?

sli.do

#Pr-QA

Diamond Partners



THE CROWN IS YOURS



Table of Contents

1. Course Objectives
2. Training Team
3. Course Organization
 - Course Infrastructure
4. Exam
 - Evaluation Criteria
5. Homework
6. Resources





Course Objectives and Scope

Course Scope



- Develop a strong foundation in **programming concepts** and **techniques**, enabling students to confidently write code and solve problems
- Understand the significance of **unit testing** and how to use it effectively to **validate code functionality** and ensure **software reliability**
- Cultivate **critical thinking** and **problem-solving skills** through various coding challenges, exercises and projects

Course Scope



- Fundamental **concepts of programming**, including data types, variables, expressions, and simple calculations
- Using **conditional statements**, such as if-else and switch-case
- Using while and for **loops**, including nested loops, for more complex problem-solving
- Working with **arrays** and **lists** for efficient data handling and processing

Course Scope

- Understanding the concept of **methods** and how to write and call them
- Essential knowledge of **unit testing** and using **NUnit** to verify the correctness of the code





Training Team

- Top Student @ **SoftUni**
- Training Delivery @ **SoftUni**
- Enjoys working with **C#** and **.NET** technologies, and more recently, with **GO**



- **Top Student @ SoftUni**
- Completed the **C# track @ SoftUni**
- Master's degree in Economics from UNWE
- **Senior Software Engineer @ coBuilder**





Course Organization

Course Timeline

6-Jan-2025

2-Mar-2025

15-Apr-2025

Programming Fundamentals and Unit Testing

8 weeks * 3 times / week

12 credits

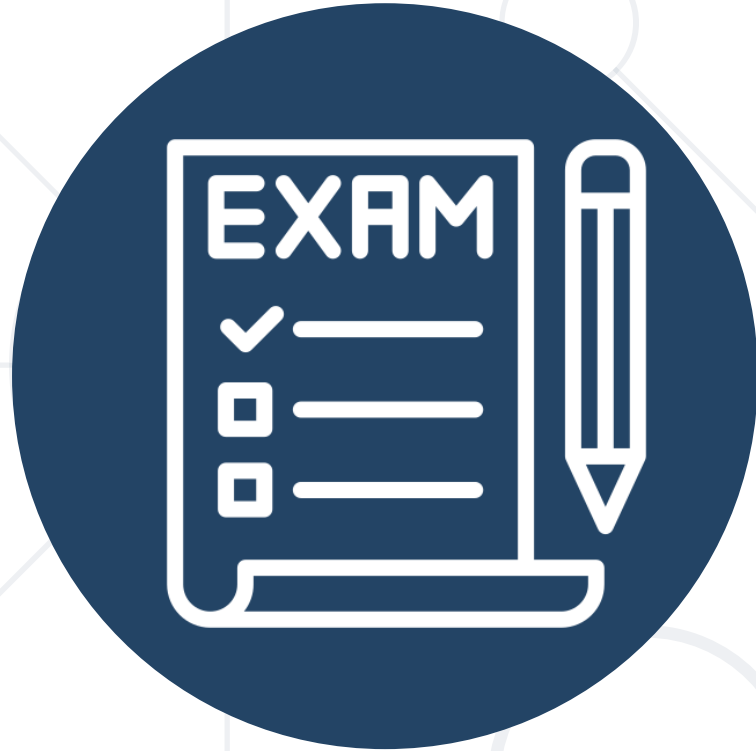
Start: 6-January-2025

Theoretical Exam: 2-March-2025

Practical Exam: 2-March-2025

Theoretical Exam Retake: 15-April-2025

Practical Exam Retake: 15-April-2025



Exam

Practical Exam Overview

- Practical exam tests the following skills:
 - **Loops**: Students will be tested on their ability to utilize loops, to perform repetitive tasks and handle iterations in their programs
 - **Lists / Arrays**: The exam will assess students' proficiency in working with lists and arrays, to manage data and solve list-related problems
 - **Unit Testing Methods**: Students will demonstrate their understanding of unit testing by writing test cases to verify the correctness of individual methods
 - **Unit Testing Lists and Arrays**: The exam will include testing scenarios where students are required to verify the functionality of methods involving lists and arrays to ensure overall code reliability



Theoretical Exam

- You will have **30 minutes** once you enter
 - **10** questions
 - Multiple-choice with **1** correct answer
 - English
- Automated quiz system
- Available during the **practical** exam and **30 minutes after it**
 - You can submit your answers just **one time**
 - Advice: Start it when you **finish** the **practical exam**









Homework

- Your **homework** is mainly work in class
 - **Lesson days** → slides + live demos + exercises
 - **Exercise days** → only exercises
- How to submit your homework?
 - Submit it in the **judge system**
- Do your homework when it's due
 - **Assignments pile up quickly**



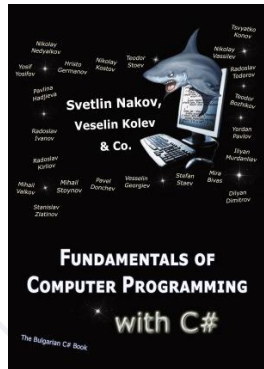


Resources

- The course assignments require to **search in Internet**
- This is an important part of the learning process
- Some exercises intentionally have no hints
- Learn to find solutions!
- Quality Assurance includes **everyday searching and learning**
- No excuses, just **learn to study!**
- QAs learn new technologies, tools, languages every day!



The Free C# Fundamentals Textbook



- "Fundamentals of Computer Programming with C#", by Svetlin Nakov & Co., 2018, ISBN 9786190007784
- English and Bulgarian versions (as PDF, ePub, ...)
- Freely downloadable from:

www.introprogramming.info

Course Web Site and FB Group

- Official web site:

<https://softuni.bg/trainings/4855/programming-fundamentals-and-unit-testing-january-2025>

- Official Facebook group:

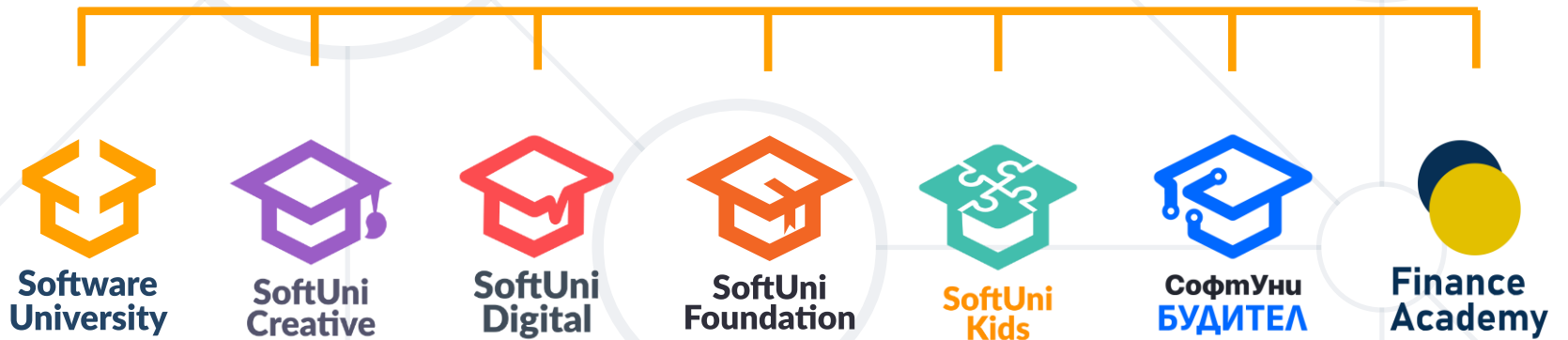
<https://www.facebook.com/groups/programmingforqajajnuary2025>



Questions?



SoftUni



- Software University – High-Quality Education, Profession and Job for Software Developers
 - softuni.bg, about.softuni.bg
- Software University Foundation
 - softuni.foundation
- Software University @ Facebook
 - facebook.com/SoftwareUniversity



- This course (slides, examples, demos, exercises, homework, documents, videos and other assets) is **copyrighted content**
- Unauthorized copy, reproduction or use is illegal
- © SoftUni – <https://about.softuni.bg/>
- © Software University – <https://softuni.bg>

